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AI-powered mobile-based early screening system for learning difficulties in children using deep learning and machine learning

Minahil Naseer¹, Hamna Kashif¹, Kashif Sultan¹, Hina Ghafoor², Awais Majeed¹ and Prince Semba Yawada^{3*}

*Correspondence:

Prince Semba Yawada

yawadaprinced@gmail.com

¹Department of Software Engineering, Bahria University, Islamabad, Pakistan

²Department of Professional Psychology, Bahria University, Islamabad, Pakistan

³Higher Institute of Science and Technology, University of Bangui, Bangui, Central African Republic

Abstract

Learning disabilities such as dyslexia, dyscalculia, and dysgraphia tend to remain undiagnosed during early childhood owing to a lack of standardized assessments, requirement of specialized professionals to conduct them, and the complexity involved in diagnosis. This research focuses on designing an early screening tool with the help of Artificial Intelligence and a mobile application that aims to serve as the initial marker for assessing whether there are any issues related to reading skills, mathematical abilities, or handwriting among 5-7-year-old children. There are three main modules included in the proposed system: (a) arithmetic module that relies on quizzes to identify the challenges associated with dyscalculia; (b) handwriting analysis module with deep learning algorithm and a VGG16-based convolutional neural network to detect possible risks of dysgraphia; and (c) reading module that utilizes games to analyze phonological problems related to dyslexia. The data consists of 395 handwritten samples obtained from several educational institutions supervised by teachers. The handwriting and mathematical modules are experimentally validated, while the reading module is currently implemented as a prototype. Experimental results obtained from a stratified train/test split (80/20) and cross-validation indicated very high predictive power for dysgraphia (accuracy rate: 91%) and dyscalculia (accuracy rate: 94%). The reading component is available in the form of a prototype that has not been subjected to quantitative analysis yet; nevertheless, behavioral testing clearly indicated differences between normally developing children and children who had issues related to phonological processing and sequencing. Even though it cannot be considered a diagnostic tool, this system represents a practical and entertaining way of identifying children who might have learning problems, which may help parents and educators recognize when professional help should be sought out.

Keywords Learning disabilities, Machine learning, Deep learning, Dyscalculia, Dysgraphia, Dyslexia, Convolutional neural network (CNN)



1 Introduction

Specific learning difficulties (SLDs), including dyslexia, dyscalculia, and dysgraphia, are neurodevelopmental disorders characterized by cognitive impairments that affect learning-related abilities [1]. These disorders manifest as persistent difficulties in reading (accuracy, fluency, and comprehension), numerical reasoning, and handwriting (including spelling and grammatical accuracy), despite adequate instruction and support at home or school [1]. When left unidentified during the early primary years, SLDs can lead to sustained academic challenges, reduced self-esteem, behavioral withdrawal, and long-term educational disadvantages.

Impairments in phonological awareness, reading comprehension, and rapid automatic naming have been identified with regard to dyslexia and affect the ability to fluently read. Dyscalculia is related to problems in number sense, arithmetic operations, and quantitative reasoning. In case of dysgraphia, issues related to poor writing skills, including problems with letter writing, spacing, and coordination are observed. Even though the above disorders have been extensively researched, many people still fail to understand dysgraphia correctly, thinking that it can be explained through bad habits of writing or laziness on the part of the child. However, like other disorders, dysgraphia requires prompt attention [2].

However, traditional approaches in diagnosis require the use of standard psychometric tests conducted by professionals. Although such methods are valid from a clinical perspective, they may be unaffordable in low-resource environments, owing to the cost involved, long referral process, scarcity of trained personnel, and parental ignorance. Consequently, many students go without diagnoses until the secondary level of education.

The recent improvements in artificial intelligence (AI) provide promising opportunities for the development of automated detection systems that will be able to assess the learning behavior of children through handwriting, reading, and number-related activity patterns. It is already proven that machine learning algorithms are able to detect abnormal reading behavior from textual, spoken, and linguistic cues [3, 4]. Also, there exists a platform called Dytective, which emphasizes the benefits of using gamified interfaces for early diagnosis of dyslexia [5]. Handwriting pattern analysis is currently a popular research area, where deep learning algorithms such as CNNs (VGG16 and ResNet in particular) are used for detecting abnormalities in motor coordination and stroke formation [6]. Machine learning algorithms have also demonstrated impressive results for classification of number-related activities [7].

However, these innovations come with their own limitations, as most of these systems either concentrate on one mode of input or need to be conducted in controlled settings. Furthermore, there is a scarcity of methods that can offer combined and multidisciplinary screening in a single system [8]. It should be noted that some research has been conducted using artificial data or experimental settings instead of actual classroom settings [9].

To address these disparities, this study suggests an AI recognition system in the form of a mobile application for children aged between 5 and 7 years, integrating three modules as follows:

- Handwriting analysis module using VGG16 convolutional neural network algorithm trained on handwriting samples in the real world.
- Numerical analysis module in the form of quizzes that employ machine learning classifiers to diagnose patterns related to dyscalculia.
- Interactive reading module used to measure phonological awareness and sequencing response patterns linked to dyslexia.

The objective of this research is not to conduct a clinical assessment but to develop an easily accessible early-screening tool that will enable parents and teachers to detect children requiring further diagnostic tests.

Contributions of this work are summarized below. First, a unified mobile-based screening approach is developed that considers the assessment of dyslexia, dyscalculia, and dysgraphia in a common platform, overcoming issues related to fragmented approaches that are being used now-a-days. Second, a deep learning-based handwriting assessment module using the VGG16 CNN algorithm based on real classroom scenarios is incorporated into the designed system. Third, the proposed design includes behavioral task-based assessment and machine learning-based approach for assessment of learning disorders, thus supporting multimodal evaluation. Fourth, the proposed solution is designed for deployment in resource-constrained environments with the objective to develop an accessible, scalable, and user-friendly screening tool through mobile applications. Fifth, the developed solution is evaluated experimentally with respect to its handwriting and mathematical components whereas the reading component of the proposed solution was recognized as a prototype. It should be noted that although the handwriting and mathematical modules are experimentally validated, the reading module remains at prototype level and is designed to be developed further in the future.

2 Literature review

Learning disabilities have recently been easier to detect owing to the development of new technologies that rely on artificial intelligence and machine learning algorithms. Currently, research has largely been devoted to three important issues related to learning disabilities: dyslexia, dyscalculia, and dysgraphia, each one presenting a unique view to the use of automated detection. In this part, an extensive review will be provided of existing research in relation to the above-mentioned learning disabilities.

2.1 Computational approaches for dyslexia screening

A considerable number of studies have been conducted on the automated detection of dyslexia by using language analysis, speech processing, and reading behavior analysis. In [3], Gómez et al. utilized automatic speech analysis approaches to detect phonological errors, hesitation phenomena, and prosody anomalies in children's reading process. Their findings suggest that acoustical characteristics could be useful for predicting reading problems at an early stage. A study conducted by Rello and Ballesteros [4] demonstrated how a machine learning algorithm based on linguistic characteristics was able to classify dyslexic readings with an accuracy rate exceeding 80%. Later, Rello et al. [5] developed Dytective, a gamified reading website that combines linguistic characteristics with behavioral performance in the task.

In addition to text and speech analysis, game-based digital platforms have shown promise as tools for evaluating phonological awareness and working memory. Recent

studies indicate that interactive reading tasks can effectively detect difficulties in decoding, sequencing, and rapid automatic naming—key indicators of dyslexia risk [6]. However, despite their potential, many of these systems require controlled environments or are designed for older children, limiting their applicability in early primary classroom settings.

2.2 Machine learning models for dyscalculia screening

Dyscalculia, described by determined difficulties in numeric processing and computing performance, has been examined using behavioral data and machine learning. Later studies show limitations in analyzing symbolic magnitudes, mental calculation, and number line assessments as key behavioral characteristics. Multiple machine learning classifiers, such as logistic regression, random forests, and support vector machines, have attained high accuracy in differentiating between typical and at-risk math students [7].

Further studies show the range of computational tasks based on surveys or games for recording actual behavioral patterns in terms of precision, reaction time, and response reliability. These models enable adaptable and unobtrusive screening without clinical assessment tools. However, present systems rely on remote computational tasks and are not incorporated into extensive, integrated screening applications [10].

2.3 Deep-learning approaches to dysgraphia and handwriting difficulty detection

The detection of handwriting disorders has improved reasonably from advancements in deep learning models, especially convolutional neural networks (CNNs). Prevalent research depicts that CNNs can obtain detailed spatial features, such as stroke consistency, curvature, spacing, and letter patterns, which are tough to capture manually. The VGG16, ResNet50, and Inception architectures have been accomplished and are now used to classify handwriting samples with dysgraphia-like characteristics [11].

Studies performed on datasets of handwritten samples collected using tablets, digital media, or controlled settings have been able to demonstrate excellent accuracy in classification. However, there are limited studies conducted on the application of images captured from a cell phone in real-life classroom settings. Such studies pose various difficulties like varying light intensity and irregular surface for writing. The above research gap motivates the implementation of the handwriting module in the proposed study.

2.4 Integration screening systems and research gaps

Although substantial development has been made in all three areas, the literature identifies several limitations that restrict practical application:

- Fragmentation of tools: various prevalent solutions pivot on a single learning difficulty rather than providing a united, multi domain assessment system.
- Requirement of specialized hardware: Speech evaluator, digitizing tablets or neurocognitive testing gadgets are often needed, limiting scalability.
- Limited early-age validation: Many researches rely on grown up children rather than the critical 5–7-year age when learning trends are most adaptable.
- Lack of mobile-based, classroom-feasible tools: Few findings work purely with conventional smartphones, making them unreachable in low resource settings.

- Scarcity of ecologically valid datasets: Many findings rely on laboratory recordings or stylus based handwriting data, which may not depict actual classroom performance.

These void spaces draw attention towards the need for a reachable, child-friendly mobile solution skilled at capturing initial indicators of behavior in reading, handwriting, and arithmetic, which encourages the present study. Table 1 provides a comparative analysis

Table 1 Summary of reviewed literature

Refs. Year	Target disorder	Methodology	Dataset type	Reported performance	Key limitations
[5] 2016	Dyslexia	Developed Dyetective, a game-based dyslexia detection tool.	Behavioral game data	Moderate accuracy	Limited gameplay duration; lacks deep modeling
[6] 2024	Dyslexia	Review of AI-based dyslexia detection techniques	Multiple datasets	Not applicable	Mostly; lacks real-world application examples.
[11] 2020	Dysgraphia	Machine learning-based dysgraphia	Handwriting datasets	High accuracy	Limited generalization due to dataset constraints
[12] 2009	Dyscalculia	Explores co-occurrences of developmental disorders like dyscalculia	Experimental datasets	Not reported	Focuses on dyscalculia without integrating dyslexia or dysgraphia
[13] 2019	Dyslexia	Uses neural networks to detect dyslexia from handwriting	Handwriting images	High accuracy	Limited dataset; lacks real-time analysis
[14] 2019	Dyslexia	Automated dyslexia detection using handwriting images	Image dataset	Moderate–High accuracy	Focus only on handwriting, ignoring phonics and reading aspects.
[15] 2019	Multi-domain	Deep learning-based screening for dyslexia, dysgraphia, and dyscalculia	Mixed datasets	High accuracy	Limited interpretability of deep learning models
[16] 2017	Dysgraphia	Handwriting analysis for dysgraphia detection	Stylus/digital handwriting	High accuracy	Focuses on handwriting, ignoring other detection aspects
[17] 2019	Dyslexia	Machine learning-based dyslexia detection via eye-tracking	Eye movement dataset	High accuracy	Requires specialized hardware for eye-tracking.
[18] 2010	Dyslexia	Genetic fuzzy systems effectively detect dyslexia using low-quality and noisy data.	Controlled datasets	Moderate accuracy	The method depends on hand-crafted features and controlled datasets.
[19] 2019	Dyslexia	Machine learning using eye-movement features can identify children with dyslexia	Eye-tracking data	High accuracy	The method requires specialized eye-tracking equipment
[20] 2020	Dyslexia	Repeated sight-word exposure improves reading mastery in at-risk first graders.	Educational datasets	Not applicable	The study does not address automated or large-scale screening
[21] 2023	Dyslexia	A game-based approach can be used to screen for dyslexia in an engaging way	Prototype dataset	Prototype-level	The system is evaluated only at a prototype level
[22] 2024	Dysgraphia	VGG16 with transfer learning achieves high accuracy in offline handwritten words.	Offline handwriting dataset	High accuracy	The study focuses on script recognition

of existing approaches in terms of methodology, dataset characteristics, and reported performance, enabling a clearer identification of research gaps.

The comparison in Table 1 reveals clear differences across methodologies, datasets, and reported performance. While deep learning and eye-tracking-based approaches often achieve high accuracy, they typically rely on controlled datasets or specialized hardware, limiting real-world applicability. In contrast, game-based and behavioral approaches are more accessible but often lack robust validation. Additionally, most studies focus on a single learning disorder, highlighting the lack of integrated multi-domain solutions. These observations underline the need for a unified, mobile-based screening system, which the proposed work aims to address.

3 Methodology

This section provides detailed explanation concerning recruitment of participants to this research project, the procedure used in data collection, formation of the dataset, its pre-processing, development and evaluation of models for an early warning system proposed.

3.1 Participants and study setting

Participants in the study were children from age 5 to 7, who are considered to be at primary education level where children develop skills in reading, writing and mathematics. Participation was voluntary and needed consent from parents or schools.

- Handwritten samples were collected from different educational settings, namely:
- APS special school.
- Two branches of Dar-e-Arqam.
- APS ordnance road campus.
- Cantt Public Girls High School & College, Sabzazar.

The data were collected in a controlled classroom environment under teacher supervision to ensure consistency.

While the dataset size is moderate, it consists of real-world samples collected from educational institutions, providing practical variability and ecological validity. To mitigate dataset limitations, data augmentation techniques were applied, and model evaluation was conducted using robust validation strategies to reduce overfitting and improve generalization. The dataset exhibits class imbalance, which was addressed through appropriate sampling strategies and augmentation techniques to ensure balanced model learning and fair performance evaluation.

The dataset was evaluated using a stratified train–test split (80/20). Additionally, k-fold cross-validation was applied to the machine learning models to improve robustness and reduce bias associated with limited data. Cross-validation allows the model to be evaluated across multiple data partitions, improving reliability of the reported results. It is important to note that the dysgraphia and dyscalculia modules are experimentally validated using collected datasets, while the reading module is currently implemented as a prototype and has not yet undergone full quantitative validation.

3.2 Inclusion and exclusion criteria

To avoid ambiguous classification of participants, one of the main concerns of the experts was that the following criteria were applied:

3.2.1 Inclusion criteria

- Admitted in preschool, kindergarten or early primary grades.
- Normal or corrected vision and hearing.
- Able to comprehend basic task instructions.
- There is no prior diagnosis of neurological, motor or developmental disorders (except for possible learning difficulties reported by the teachers).

3.2.2 Exclusion criteria

- Uncorrected sensory impairments.
- Known neurological disorders (e.g., epilepsy or motor disabilities).
- Non-native speakers of the language of instruction (to avoid confusion with reading tasks).
- Incomplete or inconsistent data submissions.

In this study, no child was labeled as “dyslexic,” “dyscalculic,” or “dysgraphic.” Instead, the reported labels reflect task-specific performance difficulties rather than clinical diagnoses of specific learning disorders, thereby avoiding unsupported diagnostic conclusions.

3.3 Dataset composition

Initially, a total of 395 handwriting samples were collected:

- 38 samples from children with teacher reported handwriting complication (APS Special School and Dar-e-Arqam).
- 357 samples from general classrooms (typical performance).

Because the dataset was not balanced, the additional data extension was only applied to the training dataset to guarantee that no information entered the test dataset.

After increment, the final handwriting dataset consisted of 1,069 images across two performance classes:

- Handwriting difficulty.
- Typical handwriting.

Math and reading modules were enforced and tested as functional specimen; behavioral data was recorded but not yet used for large scale modeling because of the scarce size of the initial sample.

3.4 System overview and application framework

This section furnishes the general framework of the proposed AI-based mobile application, designed for the early detection of reading, writing, and math challenges in children aged 5 to 7. The system aims to provide an integrated platform capable of capturing behavioral indicators in three learning areas using a standard smartphone, without requiring dedicated hardware or clinical trials.

3.4.1 System objectives

The application originated with four predominant objectives:

1. Accessibility: authorizes initial screening using affordable, broadly accessible mobile devices.
2. Multi-domain Assessment: incorporating reading, handwriting and numerical components within a unanimous platform.
3. Child-friendly Interaction: use easy, appealing tasks best suited for young learners.
4. AI-Driven Interpretation: examines captured behavioral data using machine learning and deep learning models to generate early risk indicators.

The system is deliberated as an introductory screening tool rather than a diagnostic instrument.

3.4.2 Module-level functional overview

The reading screening module offers the users brief, interactive reading tasks specifically designed for observing initial reading patterns including phoneme awareness, sequencing, and sight words recognition. The children will perform those activities by simply touching and hearing, giving the opportunity to the computer system to detect reading-related issues.

The handwriting screening module lets the child write common words on paper, which will be recorded by using the camera installed in the mobile device. By analyzing the written letters and words, potential problems with handwriting can be detected.

Finally, the mathematical screening module includes brief tasks related to arithmetical skills and abilities in the form of quizzes. The performance of the child in terms of accuracy and behavior will reveal mathematical reasoning problems.

3.4.3 Rationale for an integrated mobile-based system

Previous studies have shown the finest accomplishments of screening methods for reading, writing, and mathematical tasks in discrete domains; however, solutions accessible for dissimilar domains remain restricted. The proposed approach offers various advantages:

- Unified platform: Lessens split and enhances usability.
- Mobile based implementation Increases reach in low resource educational contexts.
- Real world data integration: Handwriting samples composed from classrooms enhances ecological validity.
- Modular expansion: Enables addition of new tasks or updated models in future releases.

By uniting three learning domains, the system aligns with comprehensive educational screening practices and supports early recognition pathways.

3.4.4 Ethical and practical considerations

The system is specifically designed as a reinforcement detection tool, not as a diagnostic device. To ensure responsible use:

- Results are presented qualitatively i.e. low moderate and elevated difficulty indicators.
- No clinical labels are assigned.

- Users are motivated to seek professional evaluation when applicable.
- All data are anonymized, with no storage of personally recognizable information.
- Contribution is discretionary and supervised by an adult.

These considerations directly address experts concerns regarding ethical clarity and the exaggeration of claims.

In terms of technology, the presented system is based on an elaborate pipeline of structured data processing, whereby user interaction is captured via the mobile interface and subsequently converted into structured inputs for further inference of the models. In particular, handwriting images will undergo preprocessing and subsequent analysis with the use of a convolutional neural network; numerical and reading task responses will be coded as features and subsequently examined by means of machine learning algorithms.

The system has been designed to work under realistic conditions in classrooms, where various limitations such as the lack of computing power, varying lighting, and heterogeneity of devices can be encountered. Therefore, effective preprocessing algorithms and models have been chosen to ensure timely and responsive interactions even on common smartphones. The system is not dependent on continuous internet access, thus providing the opportunity for its deployment in educational settings with limited computing infrastructure.

3.5 Mobile application design and user interfaces

The mobile application has been developed for the purpose of early screening for problems with reading, writing, and math among children between the ages of 5–7 years. The user interface of this application has been designed not only with usability in mind but also with reliability in mind because the way tasks are presented can have an impact on the reliability of the behavioral inputs collected. The use of this application will require supervision by teachers/parents.

3.5.1 User interface design principles

Three major principles are considered while designing the mobile application:

- **Simplicity:** Instructions for tasks are simple to understand visually, have minimal text, and contain audio signals to assist younger participants.
- **Engagement:** Colors, icons, and gaming components are used in the interface to keep the participant engaged and mitigate any anxiety associated with the task.
- **Consistency:** Consistent layout and interactions are used in all three modules, reading, handwriting, and mathematics.

Figure 1 shows the main screen from which the user can choose the desired module.

3.5.2 Dyscalculia screening interface

The dyscalculia module contains number-related activities such as addition, subtraction, number comparison, and counting. The activities are designed in such a way that the essential performance metrics, which include correctness, reaction time, number of trials, and difficulty progression, can be measured. The data obtained from these activities is then converted into structured feature vectors that feed machine learning-based risk assessment. Figure 2 depicts the user interface of the dyscalculia module.

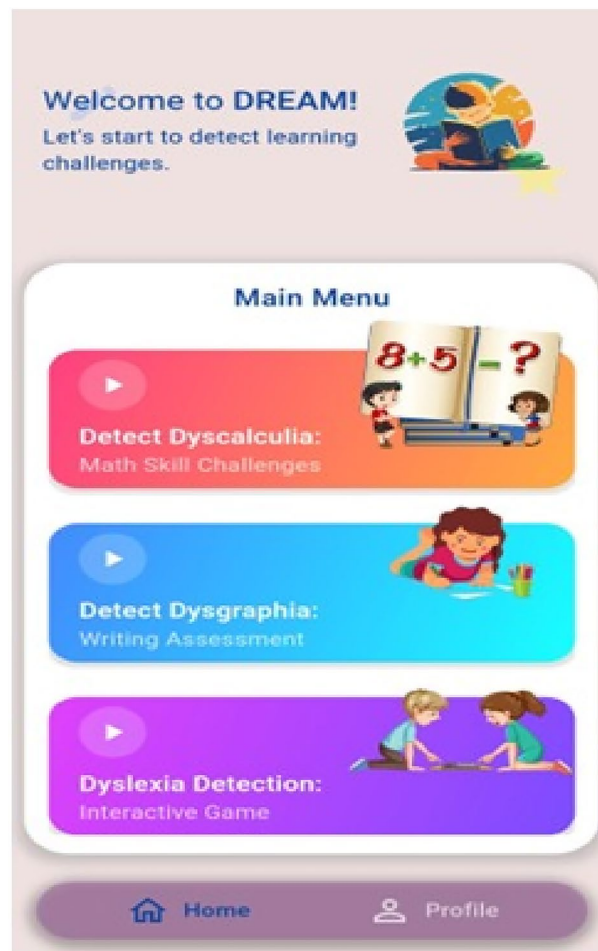


Fig. 1 Main menu of the mobile application displaying module-selection options for reading, handwriting, and arithmetic screening

3.5.3 Dysgraphia screening interface

The dysgraphia module uses the device camera to capture handwriting images. Children write predetermined words on the paper, and the images are preprocessed by cutting, converting to grayscale, removing noise, and normalizing them before passing them to the convolutional neural network. Figure 3 presents the handwriting capturing interface.

3.5.4 Reading screening interface

This module is comprised of tasks that focus on phonological awareness, word recognition, sequencing, and audio/visual integration in order to gather behavioral data linked to reading disabilities. The interface is specifically structured to avoid variability of interaction, especially since the input received while gathering behavioral data can have an impact on the results of the models' predictions.

3.6 System framework and architecture

The suggested system combines the process of mobile data gathering and uses AI logic to detect reading, writing, and mathematics difficulties among children at an early age. The entire framework can be explained using the module-based approach, which involves the components of user interaction, data processing, and AI inference. The modular

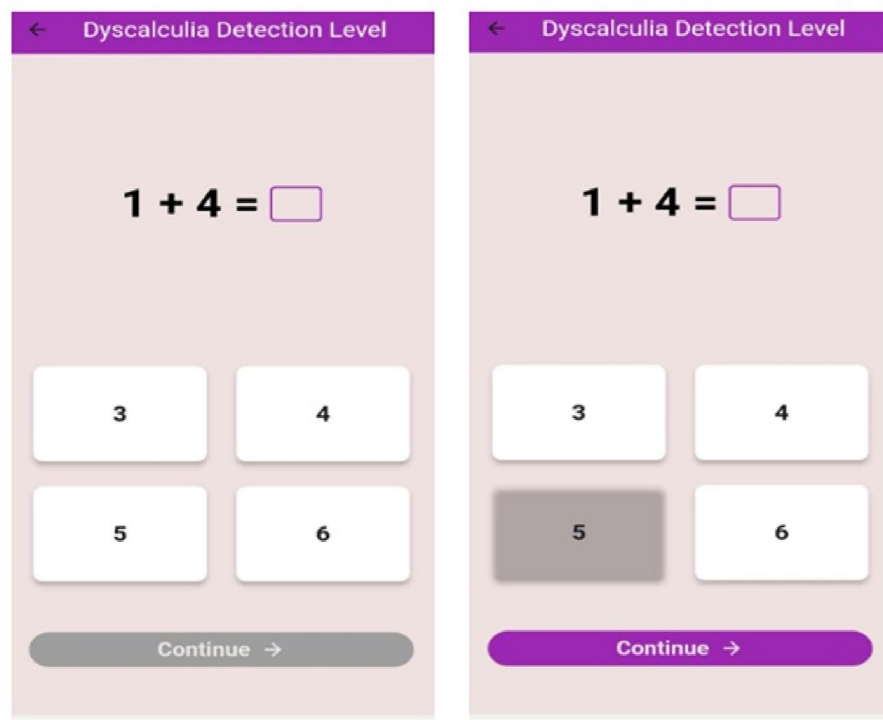


Fig. 2 User interface of the dyscalculia detection module presenting arithmetic questions and capturing behavioral performance features

architecture confirms the flexibility, maintainability, and separation of the system components. Firebase is used as the back-end service for the purpose of storing the data securely, authenticating users, and ensuring synchronization of data between the mobile application and the processing modules. Figure 4 shows the general flow and data flow in the suggested system.

3.6.1 High-level framework overview

The architecture consists of three main components that process three different types of data sources:

Handwriting Component: Handwriting samples obtained through the mobile phone's camera are first processed and then analyzed using a convolutional neural network based on VGG16.

Mathematical Reasoning Component: Numerical reasoning responses are converted into vector representations and evaluated using classification algorithms such as logistic regression, random forest, and Gaussian Naïve Bayes classifier.

Reading Behavior Component: Interaction records and timing information are used to determine behavior patterns associated with reading activities.

As shown in Fig. 4, the whole process follows a pipeline that starts from user interface, followed by data collection, pre-processing, model reasoning, and output integration to produce a consistent outcome. Every module treats their particular data separately and results obtained from these modules are then merged together to give a consolidated report. This design enhances the modularity and scalability of our approach and facilitates future addition of learning components.

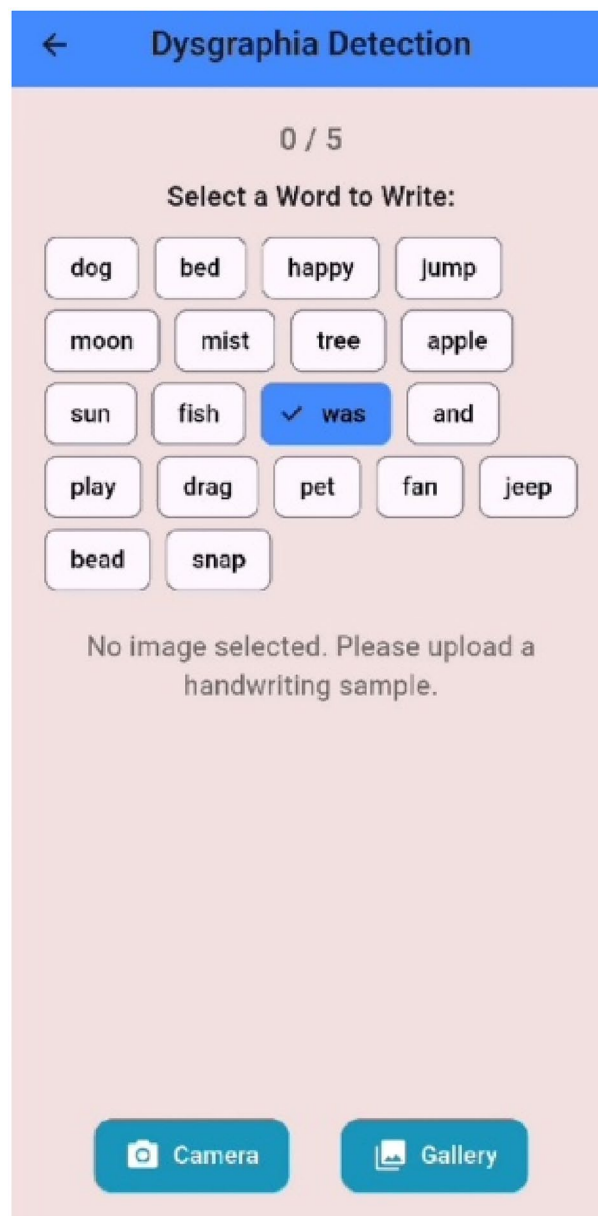


Fig. 3 User interface of the dysgraphia module

3.6.2 System architecture components

User Interface Layer: cHandles all user side interactions:

- Home dashboard.
- Module selection.
- Instructions and feedback.
- Task execution interfaces i.e. quiz, reading tasks and handwriting capture.
- Results presentation.

Data Acquisition & Preprocessing Layer: Responsible for collecting raw task data

- Handwriting images: cropped, denoised and normalized.
- Math responses: encoded into feature vectors.

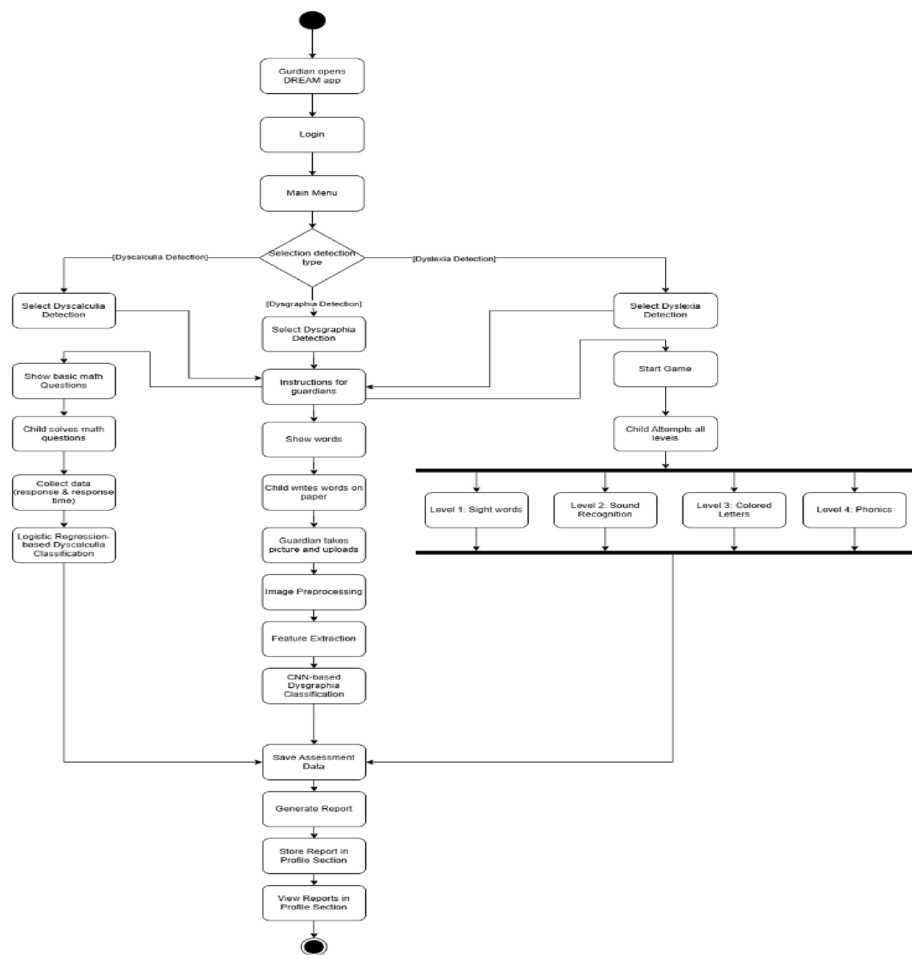


Fig. 4 End-to-end system workflow of the proposed early screening application

- Reading task behaviors: logged as event-based features.

All raw data is anonymized and pre-processed before being passed to the inference layer.

AI Inference Layer: This is the analytical core of the system:

- CNN Model (VGG16): identifies handwriting complexity indicators.
- ML Classifiers: evaluate mathematical reasoning patterns.
- Reading Behavior Analyzer: illustrate phonological and organizing behavior.

Models output qualitative threat indicators i.e. low, moderate and elevated.

3.6.3 Result aggregation layer

This layer aggregates module-level outputs to provide an overall screening summary along with domain-wise risk scores aggregated to a single output representation.

3.6.4 System performance and deployment considerations

When considering system performance, factors such as latency, memory usage, and response time are particularly important, especially for real-time mobile applications. Latency refers to the delay between a user's input and the system's response, and minimizing this delay is essential for maintaining a smooth user experience. In this work,

lightweight preprocessing methods and optimized model configurations are used to support near real-time inference. On typical mid-range smartphones, the response time is expected to be within a few hundred milliseconds. In addition, the models are designed to have a relatively small memory footprint (on the order of tens of megabytes), making them suitable for devices with limited computational resources.

From an architectural standpoint, the system is designed in a modular manner, with clear separation between user interaction, data acquisition, preprocessing, and AI-based analysis. This modular structure makes it easier to extend the system by adding new screening modules without disrupting existing components, thereby supporting future scalability and system evolution.

Compared to many existing approaches, particularly those that rely on specialized hardware such as stylus-based input [16] or eye-tracking devices [17], the proposed system is designed to operate using standard smartphones. This significantly improves accessibility and practical deployment. Moreover, while most prior work focuses on a single learning difficulty, the proposed framework brings together multiple domains within a unified pipeline, making it more suitable for real-world educational use.

In terms of implementation, the system combines mobile-based data collection with backend processing modules responsible for preprocessing and model inference. This hybrid setup allows efficient computation while maintaining the flexibility needed for future improvements and potential large-scale deployment.

3.7 Handwriting data preprocessing

The handwriting images were captured using smartphone cameras and pre-processed using a standardized process:

- Cropping: Removes background regions around text.
- Grayscale Conversion: Reduces dimensionality and color alteration.
- Noise Reduction: utilizes Gaussian blurring and verge to clean artifacts.
- Normalization: Resizes images to 224×224 pixels for VGG16.
- Pixel Scaling: Normalizes values between 0 and 1.

This method ensures consistency despite inconsistency in camera quality and classroom lighting. Figure 5 shows the handwriting data preprocessing pipeline.

3.7.1 Data augmentation strategy

To address the limited size of the handwriting dataset and improve model generalization, controlled data augmentation techniques were applied exclusively to the training set. The augmentation process included:

- Small-angle rotations ($\pm 10^\circ$) to simulate variations in writing orientation.
- Horizontal and vertical shifts (up to 10%) to account for alignment inconsistencies.
- Zooming (range: 0.8–1.2) to reflect differences in writing scale.
- Shearing transformations (up to 10%) to mimic natural handwriting distortions.
- Horizontal and vertical flipping and brightness variation to simulate real-world lighting conditions during image capture.

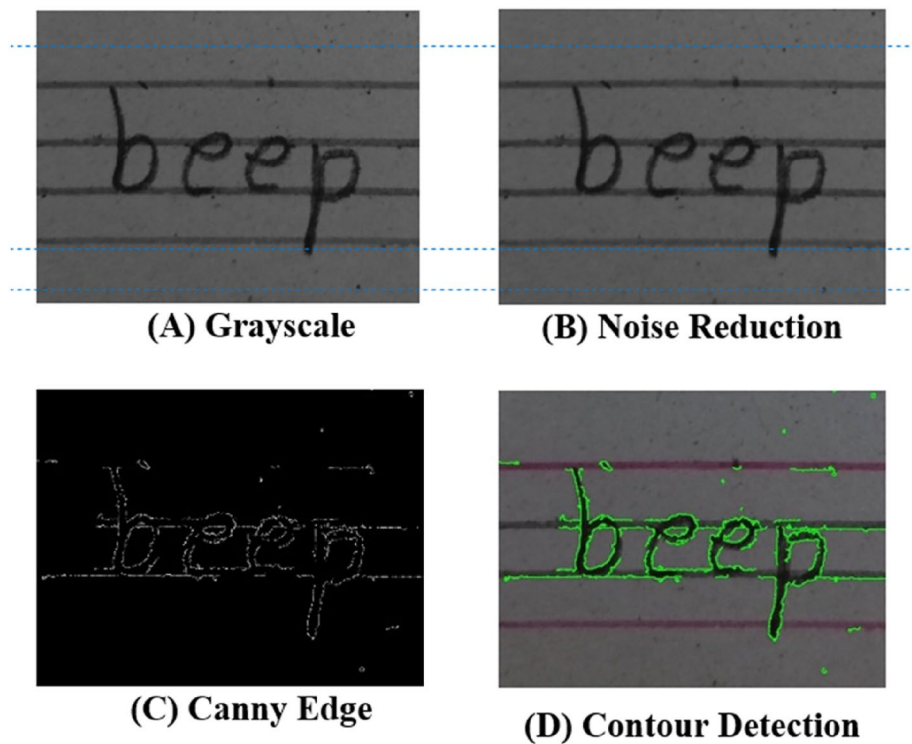


Fig. 5 Handwriting data preprocessing pipeline

These transformations were selected to preserve the semantic integrity of handwriting while introducing realistic variability. No augmentation was applied to the test set to prevent data leakage and ensure fair evaluation.

3.8 Model development

3.8.1 CNN for handwriting difficulty classification

A VGG16-based CNN is used for handwriting and character recognition. The CNN model was evaluated using a stratified train–test split, and overfitting was controlled using regularization techniques including dropout and early stopping. The model was configured as follows:

- Pretrained ImageNet weights for attribute extraction.
- Freeze lower convolutional layers.
- Custom density layers were added for binary classification.
- Adam optimizer with learning rate tuning.
- Early stopping was utilized to prevent overfitting.

The model outputs two classes:

- Typical Handwriting.
- Handwriting Difficulty Indicator.

3.8.2 Machine learning models for numerical reasoning

Three machine learning classifiers are used to analysis of quiz task performance:

- Logistic Regression (LR).

- Random Forest (RF).
- Gaussian Naïve Bayes (NB).

Input features included:

- Response precision.
- Response time in seconds.
- Number of attempts.
- Difficulty progression.
- Item wise performance vectors.
- Random oversampling was applied only to the training data after the train–test split to prevent data leakage and ensure unbiased model evaluation.

These models produce a numerical difficulty indicator.

3.8.3 Behavioral analysis for reading tasks

The reading module extracted behavioral metrics, including:

- Prioritizing accuracy.
- Phoneme identification.
- Sight word correctness.
- Latency among screen transitions.

These characteristics serve as early indicators of reading difficulties. In this version, the reading module analysis is descriptive due to the limited dataset.

3.9 Train–test split and bias control

To ensure proper data separation and avoid data leakage:

- All data augmentation was applied after splitting the data into training and test datasets.
- An 80/20 stratified split preserved class balance.
- Cross validation was applied on ML models.
- The handwriting images of the same child were not split into training and test data sets.

This ensures methodological integrity and a valid evaluation.

3.10 Evaluation metrics

Model's performance is evaluated using the following evaluation metrics:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- Confusion Matrix.
- ROC Curve and AUC (for Math module).
- Due to the limited dataset size, statistical significance testing was not performed in this study. This is acknowledged as a limitation, and future work will include larger

datasets and formal statistical evaluation to strengthen the generalizability of the results.

3.11 Ethical considerations

The methodology was developed taking into explicit account the ethical concerns raised by experts:

- Participation was voluntary in all cases.
- No clinical labels were assigned.
- The children's identities were anonymized; no personal data was stored.
- Parents and teachers oversaw every step of the data collection.
- The system furnishes risk indicators, not diagnostic labels.

The study conforms to the advised principles for early childhood educational assessment.

4 Results and evaluations

This section depicts the empirical results from the handwriting (dysgraphia) and mathematical reasoning (dyscalculia) modules, followed by the descriptive behavioral findings from the reading module. The analysis focuses on accuracy, class separation, and confusion matrix patterns to ensure transparency and avoid overly general diagnostic conclusions.

4.1 Handwriting difficulty classification (dysgraphia module)

The VGG16-based CNN was trained on a dataset of 395 original handwritten text images, which was further enhanced to 1069 images after applying a controlled data augmentation exclusively to the training dataset. The model depicted a high capacity to distinguish between indicators of writing difficulties and typical handwriting patterns.

4.1.1 Training and validation performance

During training, the model achieved stable convergence with declining loss curves and increasing accuracy curves. Figure 6 shows the loss curves for training and validation, confirming that the model did not exhibit overfitting.

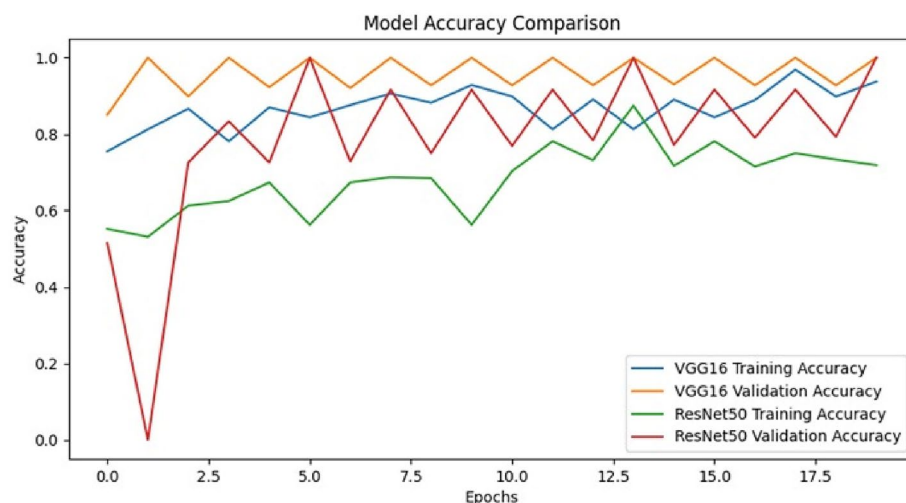


Fig. 6 Training and validation loss curves for the handwriting CNN classifier

While k-fold cross-validation is beneficial for improving model generalization, its application to deep learning models is computationally intensive, particularly in resource-constrained environments. Therefore, a train–test split combined with regularization techniques was adopted in this study.

4.1.2 Confusion matrix analysis

The confusion matrix shown in Fig. 7 shows the clear demarcation among the two handwriting performance classes.

- True Positives (Difficulty correctly identified): High.
- True Negatives (Typical handwriting correctly identified): High.
- False Positives/Negatives: Low and balanced.

This suggests that the model reliably identifies indicators of writing difficulties.

4.1.3 Classification report

The accuracy of the VGG16-based classifier is 0.91 with an equally good balance between its precision and recall at 0.94 for each category. It is noteworthy that the recall rate for the handwriting difficulty category is high, which implies a low chance of missing detections. Such findings are in agreement with previous works that have analyzed handwritten notes using deep learning architectures and are indicative of the suitability of VGG16 for handwriting difficulty detection.

Aside from the VGG16 model, the ResNet50 architecture was also developed as a baseline model for comparison purposes. Both models used ImageNet weights pre-trained without the top layers (include_top=False). The top layers included a global average pooling layer, fully connected layers (Dense 128 with ReLU activation and Dropout of 0.5), and binary classification. Initially, the convolutional layers of both architectures were frozen during training for transfer learning purposes.

The models were trained using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy loss. A consistent preprocessing pipeline was applied, including image resizing to 128 × 128 pixels, normalization, and data augmentation (rotation,

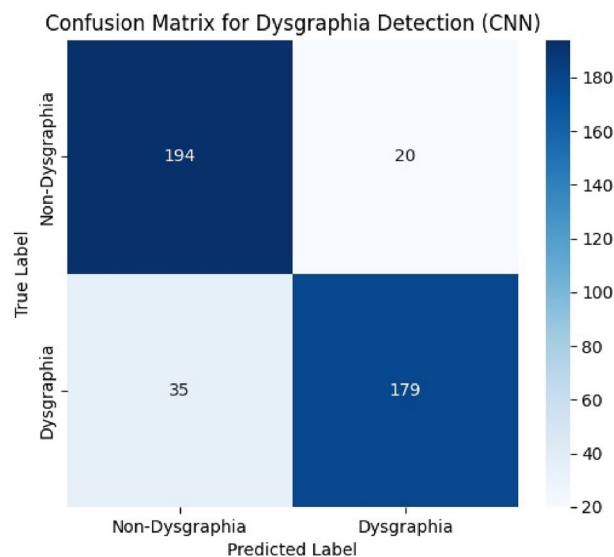


Fig. 7 Confusion matrix for the handwriting difficulty classifier

Table 2 Performance comparison of classification models for dysgraphia detection

Model	Precision	Recall	F1-Score	Accuracy
VGG16	0.94	0.94	0.91	0.91
ResNet50	0.94	0.97	0.76	0.69

Table 3 Performance comparison of classification models for dyscalculia detection

Parameter	Logistic regression	Random forest	Gaussian Naïve Bayes
Accuracy	0.94	0.98	0.72
Precision (Non-dyscalculia)	0.92	0.97	0.63
Precision (dyscalculia)	0.96	0.99	0.97
Recall (Non -dyscalculia)	0.96	0.99	0.98
Recall (dyscalculia)	0.92	0.96	0.51
F1-Score (Non-dyscalculia)	0.94	0.98	0.77
F1-Score (dyscalculia)	0.94	0.98	0.66
Support (Non-dyscalculia)	50	50	50
Support (dyscalculia)	53	53	53

shifting, zooming, shearing, flipping, and brightness variation) applied exclusively to the training set.

Both models were trained for 20 epochs with a batch size of 32 and evaluated using the same test dataset. Performance was assessed using accuracy, precision, recall, F1-score, and confusion matrix, ensuring a fair and reproducible comparison between VGG16 and ResNet50. Table 2 presents a comparative performance evaluation of CNN-based models for dysgraphia detection. Compared to ResNet50, VGG16 demonstrates superior overall performance, particularly in terms of accuracy and F1-score, indicating more consistent classification across both classes in this task.

4.2 Numerical reasoning classification (dyscalculia module)

Other than stratified train/test partitioning, k-fold cross validation was used to evaluate the performance of ML models, where the models were tested on multiple data splits to ensure robustness and reduce the biases that could result from having limited data. ML models can be biased if trained on imbalanced data sets, especially if one class is more numerous than the other. To eliminate this problem, random oversampling (ROS) was performed on the training set to have balanced learning.

Logistic Regression (LR), Random Forest (RF), and Gaussian Naïve Bayes (NB) classifiers were implemented on the numeracy test data from the questionnaire. To correct the class imbalance, resampling was done on the data before splitting and training the models using the 80/20 stratified train/test partition method.

4.2.1 Model performance

Table 3 provides the performance analysis of the three classifiers using several evaluation metrics such as accuracy, precision, recall, F1-Score, and class support. In all of these measures, the model based on Random Forest demonstrated the best performance with an accuracy of 0.98 and high precision, recall, and F1-score in both classes. It proves the effectiveness of ensemble methods in discovering trends in numerical answers to tasks. The better performance of Random Forest means that non-linear trends exist in the data.

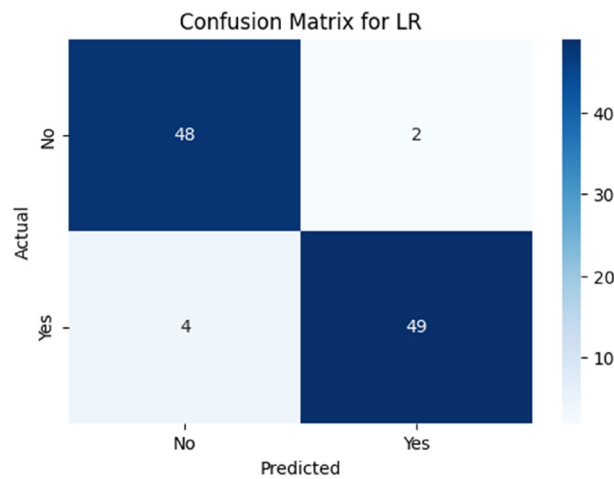


Fig. 8 Confusion matrix for logistic regression classifier

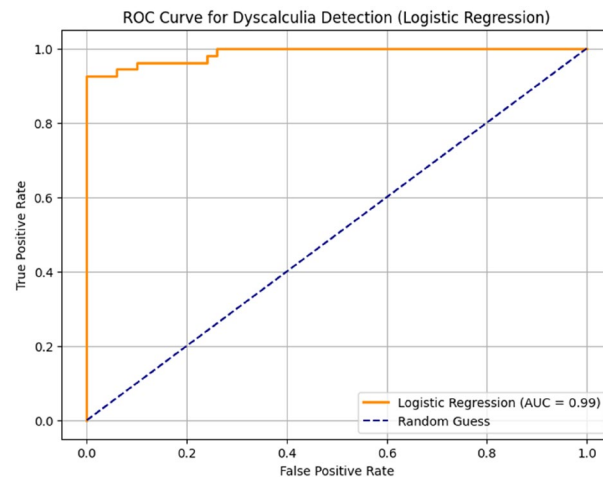


Fig. 9 ROC curve for numerical reasoning classification

4.3 Confusion matrix for LR model

The LR classifier generated a balanced confusion matrix with low misclassification rates, demonstrating stable model behavior.

4.4 ROC curve analysis

The ROC curve for LR is shown in Fig. 9. The ROC curve depicts the following facts;

- $AUC > 0.90$.
- Strong separation between classes.
- Low false-positive rate.

This supports the soundness of the numerical inference classifier.

4.5 Reading behavior analysis (dyslexia module)

The reading module is currently implemented as a prototype and therefore does not include quantitative performance evaluation. The observations reported in this module are qualitative in nature and are based on behavioral patterns observed during task

execution. Preliminary observations suggest that the module is capable of capturing relevant behavioral indicators associated with early reading difficulties. However, initial tests revealed significant patterns.

The recorded behavioral indicators included:

- Touch sequence precision.
- Time taken to complete phoneme identification tasks.
- Correct and incorrect responses in sight word tasks.
- Behavioral hesitations and re attempts.

Initial results show:

- Both Children and teacher reported reading difficulty revealed extended response times.
- Sequencing errors arises more frequently among at risk readers.
- Touch pattern logs show frequent corrections and indecision.

These trends are coherent with previous findings in the literature on reading hardness (e.g., difficulties with phonological awareness and sequencing).

Unlike the handwriting and mathematical reasoning modules, the dyslexia module was implemented as a working prototype due to the scarce availability of reading task data at this early stage of development. Consequently, no machine learning model was trained for dyslexia detection and the results shown in this section are based solely on behavioral observations and not on forecasting classification. The goal of the prototype is to verify the feasibility of the task (phonological mapping, visual word identification and sequencing) before collecting a sufficiently large dataset in future work for reliable model training.

5 Discussion

The results obtained from the proposed system demonstrate the effectiveness of the applied machine learning and deep learning models for the early detection of learning difficulties in children, specifically dyslexia, dyscalculia, and dysgraphia.

For dyscalculia detection, multiple machine learning classifiers were implemented, including Logistic Regression (LR), Random Forest (RF), and Gaussian Naïve Bayes (NB). Among these, Random Forest achieved the highest overall performance, with an accuracy of 98% and consistently strong precision, recall, and F1-scores across both classes. This indicates its effectiveness in capturing patterns in numerical reasoning data.

Logistic Regression also proved effective with good performance, achieving 94% accuracy and maintaining balanced F1-scores for both dyscalculia and non-dyscalculia classes. Its stability and interpretability allow using Logistic Regression as another practical option to deploy on a larger scale. Naïve Bayes with Gaussian kernel demonstrated weaker performance especially in recall for the dyscalculia class, showing the possibility of missing some cases. Although Random Forest achieves superior predictive performance, Logistic Regression offers a good trade-off between accuracy, stability, and interpretability, making it a practical alternative for large-scale deployment. These results should be interpreted in the context of a limited dataset.

Regarding dysgraphia detection, a Convolutional Neural Network based on the VGG16 architecture was used. The efficiency of such a model was compared to that of

the ResNet50 architecture. As a result, higher accuracy (91% vs. 69%) and F1-score (0.91 vs. 0.76) were found for the former than for the latter. The confusion matrix and ROC curves confirm this difference with the area under the ROC curve being equal to 0.94. The model demonstrated reliable results that indicate minimization of both false positives and false negatives.

The use of data augmentation played a vital role in improving the performance of CNN, particularly because of the limited size of the original dataset. Additionally, the application of handwriting-based feature extraction allowed for a meaningful representation of motor control issues in children, strengthening the dysgraphia classification.

The dyslexia detection module relied on a game-based assessment strategy that analyzed phonics, reading patterns, and visual recognition of letters or words, while not using ML model or DL model, as it is based on the heuristic approach.

6 Limitations

This study has several limitations that should be acknowledged. First, the dataset size is relatively small and was collected from a limited number of educational institutions, which may affect the generalizability of the results. Although cross-validation was applied to the machine learning models to improve robustness, external validation on independent datasets was not performed and remains an important direction for future work.

Second, statistical significance testing was not conducted due to dataset constraints. Future studies will incorporate larger and more diverse datasets along with formal statistical analysis to strengthen the reliability of the findings.

Thirdly, while the dyslexia (reading) module is only a prototype at this stage, there is no machine learning algorithm that has been trained or any quantitative assessment yet. Although there have been some interesting trends in terms of behavior in terms of preliminary observations, the lack of any quantitative validation means that the proposed multi-domain system is incomplete. In future studies, more effort will go into collecting data and validating the dyslexia model.

Fourthly, cross-validation cannot be done on the CNN handwriting module because of computation limitations. Although techniques like dropout and early stopping were employed to prevent overfitting, further validation methods like cross-validation and external validation will be explored in future studies.

7 Conclusion and future direction

To conclude, this study has introduced an efficient and accurate artificial intelligence-powered framework for detecting learning disabilities dyslexia, dyscalculia, and dysgraphia by employing a hybrid approach based on machine learning, deep learning, and gamified assessments. The proposed framework provides real-time usability and easy interpretation with minimal accessibility requirements. The modular design of the proposed system allows specific testing, including the collection of relevant data and monitoring of performance.

The proposed framework is promising; however, several aspects need improvement. Implementing real-time analytical feedback into the interface of the game-based and handwriting modules can improve their usability and make the results more comprehensible for users. Extending the list of supported languages and adjusting the questions in

accordance with specific curricula will make the proposed system more popular among educators. Integrating long-term monitoring functionality can help monitor children's development and learning trajectory over time. Furthermore, collaboration with educational institutions to collect larger and more diverse datasets will help reduce bias and improve the generalizability of the models.

Author contributions

M.N. and H.K. contributed to the conceptualization, methodology, software development, data collection, and writing of the original draft. K.S. contributed to supervision, project administration, and critical revision of the manuscript. H.G. provided expertise in professional psychology, contributed to the design of the screening framework, and assisted in the interpretation of learning disability indicators. A.M. assisted with data analysis, model implementation, and visualization of results. P.S.Y. contributed to manuscript editing, review, and final approval of the submitted version. All authors reviewed and approved the final manuscript.

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Data availability

The datasets generated and analyzed during this study contain handwriting samples collected from children and therefore cannot be publicly shared due to privacy and ethical restrictions. Anonymized data supporting the findings of this study may be available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the principles of the Declaration of Helsinki and was approved by the Ethical Review Committee of the Department of Software Engineering, Bahria University Islamabad (Approval No. BUIC/H11/SE/68809, dated 19 September 2023). Informed consent was obtained from parents or guardians of all participating children prior to data collection.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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